

OpenCHM: A MKG-based Education System for Chinese Herbal Medicine

Yushuang Yan
Northwest A&F University
Yangling, China
ysyan@nwfufu.edu.cn

Qihang Fang
Northwest A&F University
Yangling, China
fqh2024056248@nwfufu.edu.cn

Meili Wang*
Northwest A&F University
Yangling, China
wml@nwsuaf.edu.cn

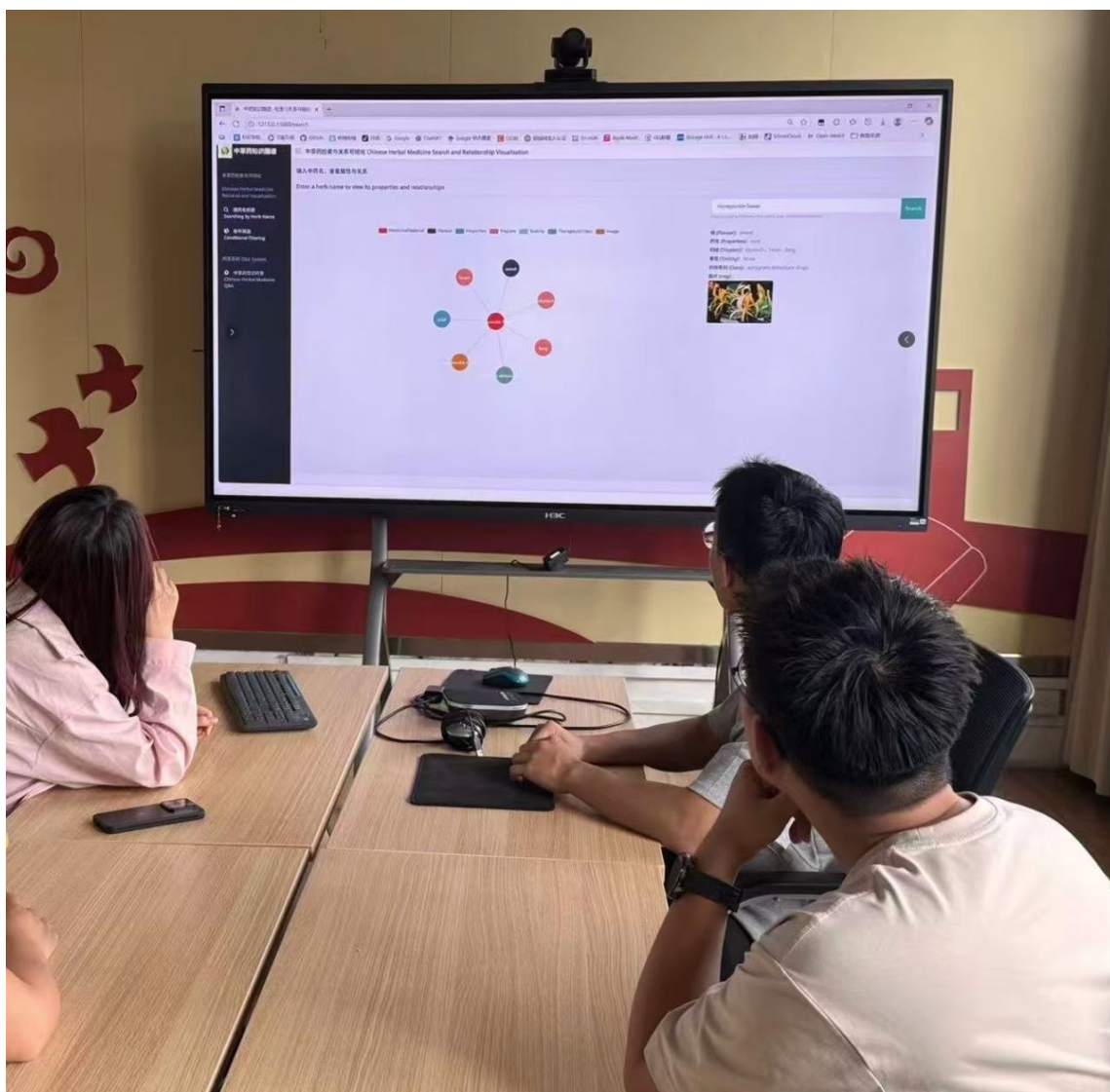


Figure 1: A full tutorial session engaging with the OpenCHM activity.

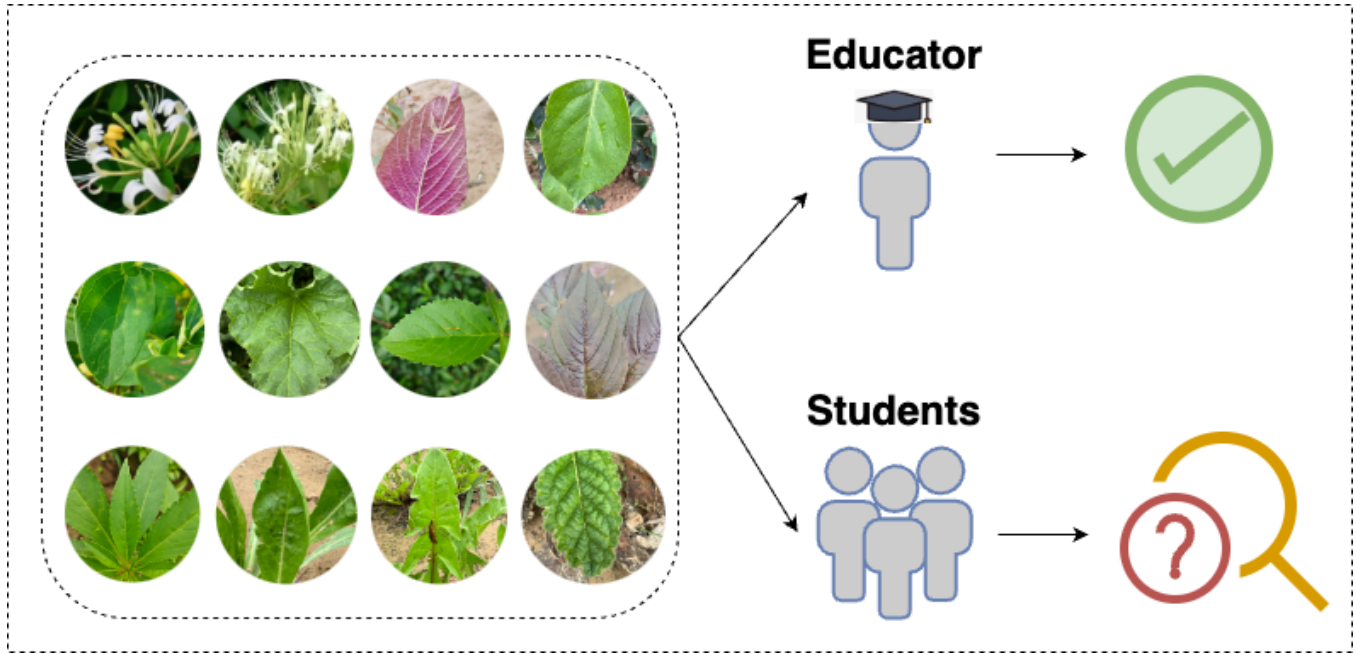


Figure 2: Students are difficult to identify the classes of CHM, particularly when CHM have similar shapes. How can they identify them quickly and accurately as the educator with specialized knowledge?

Abstract

Chinese herbal medicine (CHM), as a part of traditional Chinese medicine, is of great importance due to its remarkable performance on disease prevention, treatment, and daily healthcare. There exists a wide variety of CHM. In particular, some have similar shapes. Consequently, we face dilemmas for CHM identification. In traditional education, educators manually displaying pictures of CHM aim to insist students know the classes of CHM. This way is apparently time-consuming and labor-intensive. Knowledge graphs (KGs) are powerful and modern tools to boost model performance on various applications such as objective detection, natural language question-answering, and recommendation systems. However, existing unimodal KGs impede their potential for powerful performance improvement in downstream tasks. Nevertheless, knowledge acquisition is multimodal, including text, image, video and audio modalities. Thus, multimodal knowledge can accumulate from different views of textual and visual information. Researchers focus on multimodal knowledge graphs (MKGs). Currently, KGs used for CHM are predominantly unimodal. Due to lacking the professional knowledge, students find it difficult to gain a comprehensive

perception. Therefore, we propose the OpenCHM, a MKG-based education system for CHM, which can support image-text retrieval. Meanwhile, we construct a MKG which integrates texts and images. Our MKG can provide a wider view to explore further information of CHM. The service system is suitable for educational demonstrations as well as real-world applications. Finally, we design several MKG-based downstream tasks such as knowledge visualization, multimodal knowledge retrieval and a question-answering platform for promoting the development of CHM.

CCS Concepts

- **Artificial intelligence** → **Knowledge graph; Deep learning** ;
- **Social and professional topics** → *Computer science education*.

Keywords

Chinese Herbal Medicine, Education, Multimodal Knowledge Graph, Text, Image

ACM Reference Format:

Yushuang Yan, Qihang Fang, and Meili Wang. 2025. OpenCHM: A MKG-based Education System for Chinese Herbal Medicine. In *SIGGRAPH Asia 2025 Educator's Forum (SA Educator's Forum '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3757372.3771859>

*Corresponding author at: Northwest A&F University, Yangling 712100, China.
E-mail address: wml@nwsuaf.edu.cn (M. Wang).

1 Introduction

Traditional Chinese medicine (TCM), with a millennia-old history, is extensively utilized in human healthcare (Long et al., 2024, Zhang

et al., 2025). Chinese herbal medicine (CHM), as a crucial part of TCM, has drawn attention due to its remarkable performance on disease prevention, treatment, and daily healthcare. In particular, there are a large number of CHM as shown in Fig. 2. Some of them even have similar shapes under the same species, such as *Hansenia weberbaueriana* and *Heracleum hemsleyanum*. On the other hand, some of them in different species, may be easily classified by the global shape features while some are hard to distinguish between each other such as *Lonicera japonica* and *Lonicera macrantha* as presented in Fig 3. The classification of CHM largely relies on professionals such as educators with specialized knowledge while it is difficult for non-professionals such as students to recognize them accurately. Thus, identifying the classes of CHM poses challenges for students, which makes an obstacle for them to understand and learn about CHM deeply. In traditional education, educators manually show pictures for CHM identification, which is time-consuming and labor-intensive. Finding quick and accurate CHM identification tools are significant. Intelligent tools can improve the learning outcomes, and in the meantime bridge the gap between theory and practice. Luckily, knowledge graphs (KGs) provide opportunities to further understanding CHM from a view of linked data, making them promising tools for CHM classification. Interconnected knowledge in KGs is captured in the form of triplets <head entity, relation, tail entity>. KGs have become pivotal in a wide range of downstream tasks, including objective detection (Dewi et al., 2025), natural language question-answering (He et al., 2025), and recommendation systems (Li et al., 2024). Therefore, KGs can enhance knowledge service capabilities of CHM identification and retrieval, especially for students with non-professional knowledge.

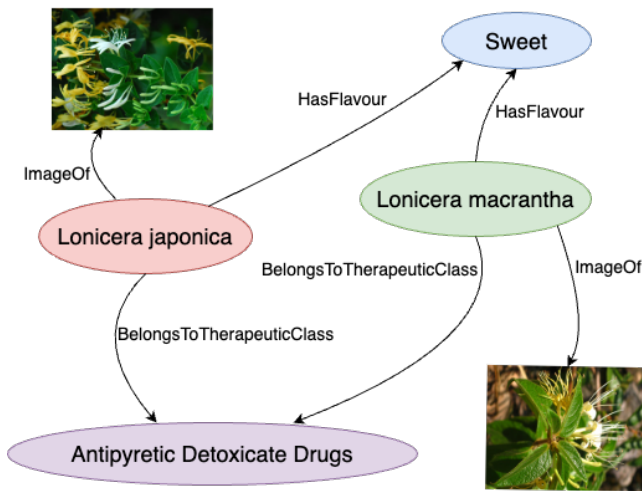


Figure 3: An example of a CHM MKG that provides textual and visual information.

We perceive our world through multiple modalities, such as vision, sound, video and audio (Wang et al., 2023) (Chen et al., 2025). Intuitively, they outperform unimodal models if models integrate data from diverse modalities due to accumulating more information. Nevertheless, recent KGs primarily draw attention to textual content and rarely incorporate multimodal information. In addition

to texts, images are also prevalent data sources. Compared with unimodal models, these various data sources can provide complementary descriptions of the same entity so that enhancing efficiency of KG-based tasks. Vision and language are two prevalent modalities when humans acquiring and expressing information. IMGpedia (Ferrada et al., 2017) is one of the first attempts to collect only image modality and then constructs a KG. (Liu et al., 2019) enriches DBPEDIA, YAGO, and Freebase-15k with numerical textual and visual information to instruct multimodal knowledge, which is an early typical instance for building MKG. MKGs have been booming in various domains from natural language processing (Chen et al., 2022) and recommendation systems (Wang et al., 2025). Researchers (Ke et al., 2025, Shen et al., 2025) have devoted to fusion of computer vision and natural language processing. However, data fragmentation impedes effective linking the CHM multimodal resources, which compromising the knowledge transmission. Current KGs solely depending on textual information can not give students a sufficient comprehensive understanding and perceptual engagement of CHM. Nevertheless, a multimodal knowledge graph (MKG) incorporates representative images and semantic information, thereby enabling students to engage in more direct cognition processes. For instance, Fig. 3 shows a simple CHM MKG which illustrates entities linking with related images. Visual information can enhance semantic expression in KG-based applications.

Thus, we propose a education system based on a multimodal knowledge graph for facilitating the understanding of CHM. The MKG incorporates visual and textual modalities so that the system can realize effective image-text retrieval due to explicit relationships between visual scenes and textual description.

Our key contributions of this work are summarized as follows:

- We develop the OpenCHM, a multimodal knowledge graph-based education system for CHM identification and retrieval, which integrates a large corpus of structured knowledge. The service system is advantageous for educational demonstrations.
- We propose a multimodal knowledge graph construction approach for CHM. The MKG can associate entities with visual data and further explore potential information from CHM images and deep semantics. We design a customized conceptual model for constructing our MKG. The proposed MKG can provide a comprehensive presentation and help students understand CHM deeply.
- Our MKG-based real-world applications such as knowledge visualization, multimodal knowledge retrieval and a question-answering platform are developed. The several key tasks can prompt effective teaching.

The remainder of this paper is organized as follows: Section 2 describes our system architecture, including data collection and preprocessing, and knowledge graph construction. Section 3 introduces real-world applications. Section 4 shows the evaluation results. Finally, Section 5 concludes our work.

2 System Overview

In this section, we describe the OpenCHM, a MKG-based system for enhancing CHM teaching and learning effectively. OpenCHM focus on modernizing teaching and simplifying the learning. Fig. 4 shows

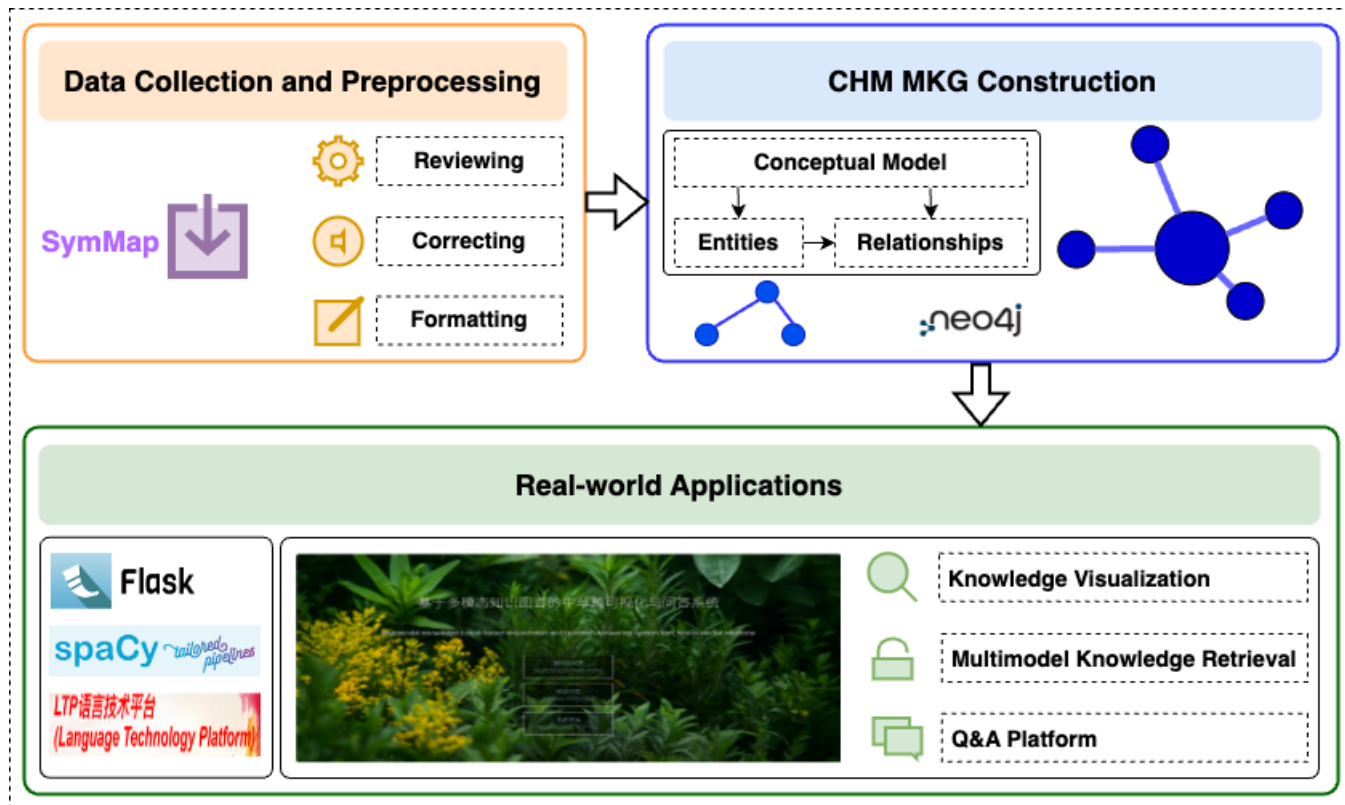


Figure 4: OpenCHM framework including data collection and preprocessing, CHM MKG construction and MKG-based real-word applications.

the OpenCHM framework. Firstly, we complete data collection and data preprocessing in Subsection 2.1. Moreover, we construct our multimodal knowledge graph in Subsection 2.2. Finally, MKG-based real-world applications are explored in Section 3.

2.1 Data Collection and Preprocessing

We collect SymMap dataset arranged in CSV files as our textual data while obtaining images from Plant Photo Bank of China (PPBC). SymMap (Wu et al., 2019) is a comprehensive database integrating traditional Chinese medicine and modern medicine, providing extensive information on herbs/ingredients and clinical symptoms. Medical experts extracted and proofread newly recorded herbs, symptoms, syndromes and their interrelationships according to the Pharmacopoeia of the People's Republic of China. From PPBC, we select one image of each Chinese herbal medicine collected from SymMap¹. After collection, the data undergoes rigorous review, correction, and formatting to generate clean and usable dataset.

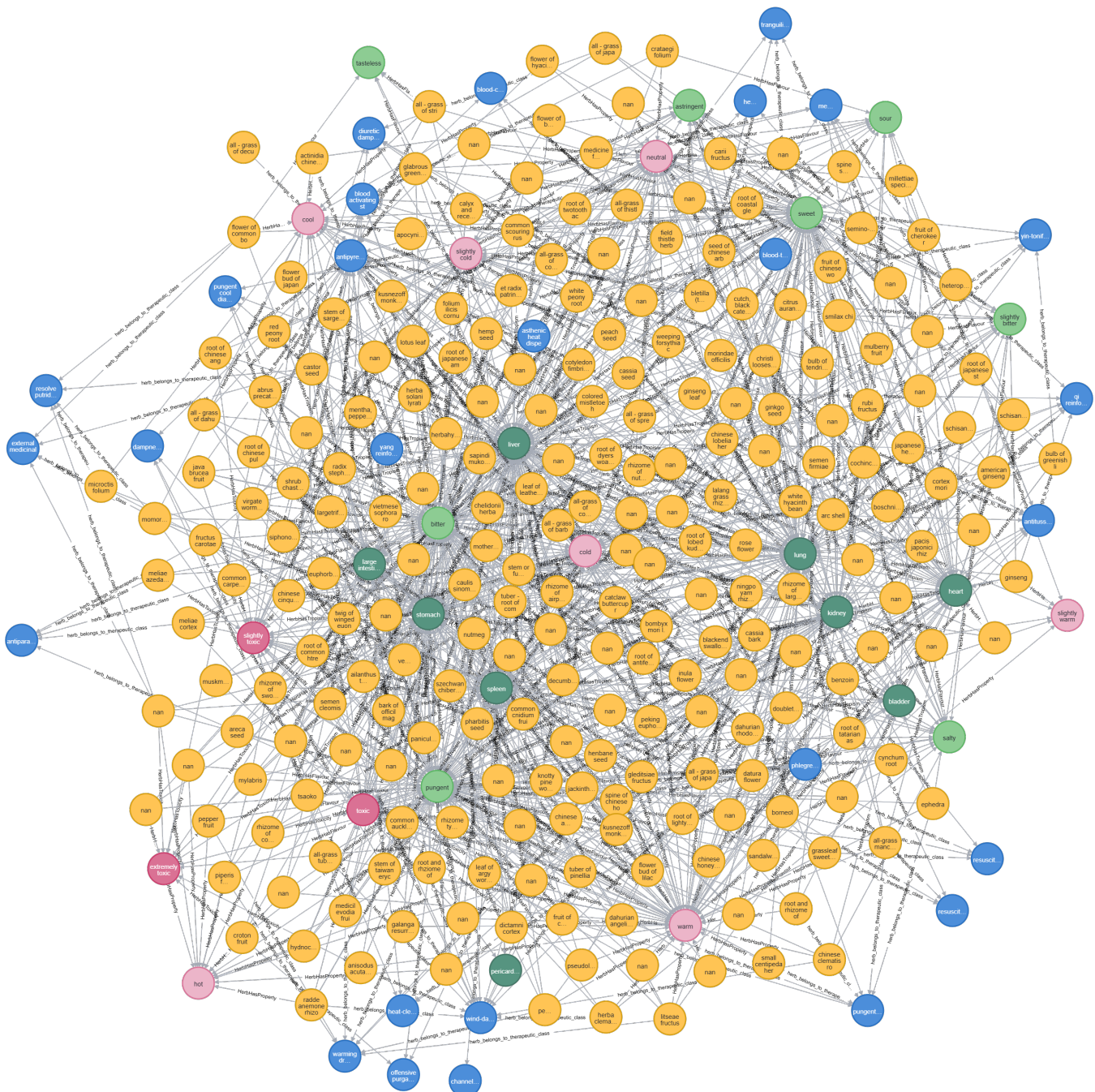
2.2 Mutimodal Knowledge Graph Construction

We aim to construct a multimodal knowledge graph based on textual and visual data resources for Chinese herbal medicine. The whole CHM multimodal knowledge graph is visualized in Fig. 5, that combines texts and images through a customized conceptual model

shown in Fig. 6. This MKG preserves the richness of CHM data while supporting advanced analysis and retrieval in modern medicine research.

Conceptual model. We carefully design a customized conceptual model as shown in Fig. 6. We try to combine the top-down and bottom-up methods for ensuring the high quality and boosting the scale of our MKG. Fig. 6 shows seven entities as nodes and six entity relationships as entity edges. From Table 1, we can see there are seven entities such as MedicinalMaterial, Flavour, Properties, TherapeuticClass, Toxicity, Tropism and Image. MedicinalMaterial shows the name of CHM. The other six entities are closely related to MedicinalMaterial (MM). Flavour refers to the taste of MM. Similarly, Properties, TherapeuticClass, Toxicity, Tropism and Image are the property, therapeutic class, toxicity, tropism and image of MM. These six entities involve subclasses. For example, Toxicity is classified into extremely toxic, toxic, slightly toxic and nontoxic. As illustrated in Fig. 6, ToxicityOf is the entity relationship between MedicinalMaterial and Toxic. For instance, ToxicityOf means *Lonicera japonica* is nontoxic when the Toxicity's value is nontoxic and MedicinalMaterial's value is *Lonicera japonica*. Named entity recognition and entity relationship extraction are the key tasks in the achievement of KGs. We use the rule-based method for information extraction. Finally, there are about 6000 entities and 5000 entity relationships in our constructed MKG.

¹The SymMap database can be accessed at <http://www.symmap.org/>.



Neo4j. Neo4j is an open-source graph database management system (GDBMS) which is a semantic network, consisting of nodes as entities and edges as various semantic relationships. In Neo4j, this graph structure provides a good way to describe complex networks and connections. Neo4j can improve efficiency in storing, querying and analyzing graph data. Compared with traditional relational databases, Neo4j is more powerful when querying complex

relationships and massive data. It is widely used in various domains, such as healthcare, recommendation systems and knowledge graph construction. Cypher is a declarative graph query language which allows for rapid querying over large-scale graph datasets. Entity relationships are visually showed after storing CHM knowledge. Thus, students can gain a more comprehensible and interactive experience.

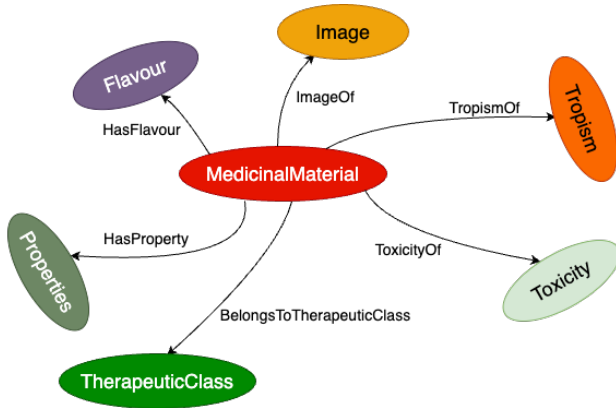


Figure 6: Conceptual model including seven entities and six entity relationships.

Table 1: Entity Category and detailed description

No.	Entity Category	Subclass	Number
1	MedicinalMaterial	<i>Lonicera japonica</i>	703
2	Flavour	bitter; salty; sweet;	1091
3	Properties	sour cold; cool; hot;	698
4	TherapeuticClass	warm qi reinforcing drugs; antipyretic detoxicate drugs; blood-tonifying medicinal	724
5	Toxicity	nontoxic; slightly toxic; toxic; extremely toxic	105
6	Tropism	heart; kidney; liver; lung	1548
7	Image	–	703

3 Real-world Applications

In this section, we develop a service system for CHM. Several MKG-based downstream tasks involving knowledge visualization, multimodal knowledge retrieval and a question-answering (Q&A) platform are explored. Consider practicality, the Q&A platform is implemented both of Chinese and English languages.

3.1 Knowledge Visualization

Knowledge graphs are rich semantic networks which can represent real-word facts. Our established MKG can bridge the gap between

fragmented knowledge of CHM. We employ interactive Flask framework and Neo4j graph database to implement CHM knowledge graph visualization. Students can get textual and visual information from the MKG. Thus, they can interactively obtain brief conceptual content. For instance, students can easily observe what the shape of *Lonicera japonica* is. Moreover, the MKG prompts students to explore potential relationships between concepts, acquiring new insights and concept. Students may discover *Lonicera japonica* and *Lonicera macrantha* have similar shapes under the guidance of educators. Next, students may consider why their shapes are similar and in what other aspects they are similar.

3.2 Multimodal Knowledge Retrieval

Our MKG for CHM can enhance the performance of knowledge retrieval. By using both of images and texts, the MKG embedded into the Q&A platform gives us a comprehensive and interactive experience. Therefore, the detailed description of the related MedicinalMaterial entity is presented if we input a name of CHM. Fig. 8 shows multimodal knowledge of Honeysuckle flower (i.e., *Lonicera japonica*), including its image and semantic relationships. Moreover, Fig. 9 shows the platform can achieve filter functionality according to five conditions. As presented in Fig. 10, CHM which fall under the heat-clearing treatment are all displayed after entering "heat-clearing".

We conclude that our MKG enables highlight core concepts and shows the structured knowledge, which enhances students to generate new views. From Subsection 3.1, the students think about whether *Lonicera japonica* and *Lonicera macrantha* have other similar characteristics. From the MKG, students can find that they possess an identical sweet flavor and fall under the identical therapeutic classification.

3.3 A MKG-based Q&A Platform for CHM

We utilize LTP and spaCy for question intent recognition to achieve text-based intelligent response capabilities. LTP² is used specially for Chinese natural language processing (NLP), and spaCy for English NLP. As shown in Fig. 7, we construct a MKG-based Q&A platform for CHM, which can use Chinese language as well as English language. The platform consists of three components: searching by a herbal name as presented in Fig. 8, conditional filtering as illustrated in Fig. 9 and Fig. 10, and Q&A as shown in from Fig. 11 to Fig. 13. Fig. 11, Fig. 12 and Fig. 13 show different examples of Q&A. Compared Fig. 13 with Fig. 9, the MKG-based Q&A allows for multi-criteria filtering, which can give students a rapid and comprehensive answer.

In summary, the Q&A platform is significantly more efficient in knowledge searching and acquisition than the human naked eyes. Students can explore various concepts, facilitate a deep understanding, and finally gain practical experience.

4 Evaluation

4.1 Experimental Setup

The study includes qualitative and quantitative experiments. The qualitative experiment recruited 32 students from Northwest A&F

²We can access LTP at <https://ltp.ai/> while accessing spaCy at <https://spacy.io/>.



Figure 7: A MKG-based Q&A platform for CHM. The platform supports several applications such as CHM searching, filtering and Q&A.



Figure 8: Multimodal knowledge retrieval. Users can input a name of Chinese herbal medicine. For example, we can get a knowledge graph of *Lonicera japonica* and its image if "Honeysuckle flower" is inputted.

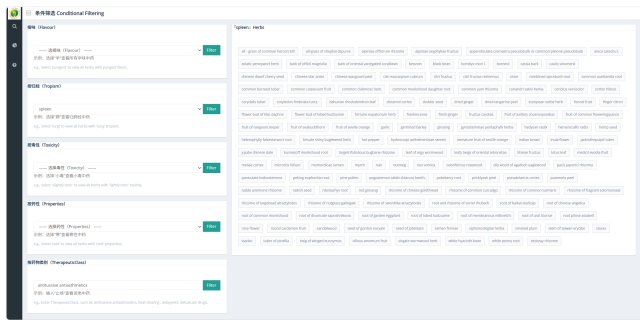


Figure 9: Conditional filtering through Flavour, Tropism, Toxicity, Properties and TherapeuticClass.

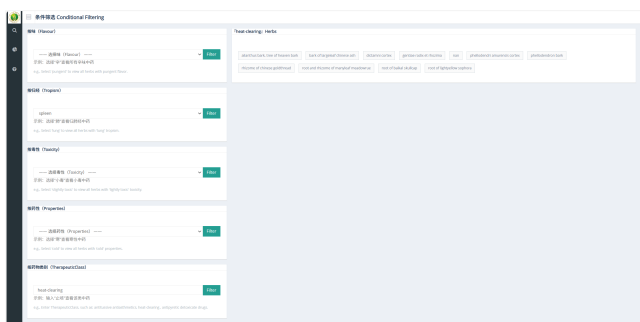


Figure 10: An example of heart-clearing filtering.

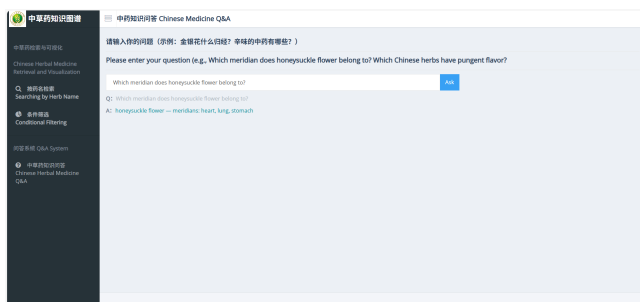


Figure 11: One of the examples of Q&A.
Q: Which meridian does honeysuckle flower belong to?

University to uncover the strengths and weaknesses of our OpenCHM through a questionnaire, in order to improve the system learning experience. The OpenCHM measured 7 dimensions within the learning experience: Knowledge Retrieval Efficiency, Interactiiviy, Information Completeness, System Stability, UI Friendliness, Real-time Response Speed and Accuracy.

The quantitative experiment recruited 30 students from Northwest A&F University to evaluate the educational effectiveness. All students were divided into 2 groups, with 15 person in each group. The first group called a traditional learning group read learning materials such as textbooks. The second group named a system learning group use OpenCHM for study.

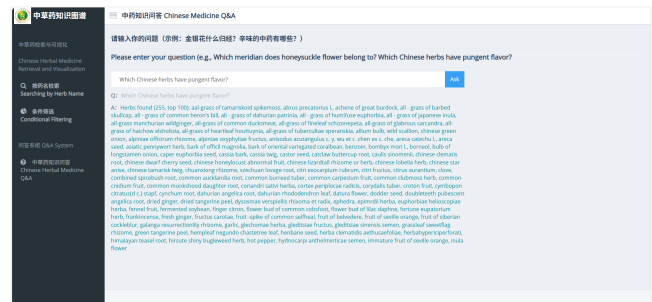


Figure 12: One of the examples of Q&A.
Q: Which Chinese herbs have pungent flavor?

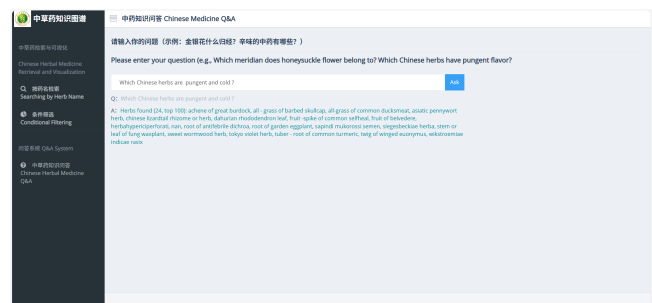


Figure 13: One of the examples of Q&A.
Q: Which Chinese herbs are pungent and cold?

4.2 Experimental Results

Fig. 14 shows the qualitative results. The majority of students are concentrated in the "Satisfied" and "Somewhat Satisfied" segments. In particular, Students are highly satisfied with Knowledge Retrieval Efficiency, Information Completeness, UI Friendliness and Accuracy. However, the number of "Not Very Satisfied" and "Dissatisfied" students are likely the highest of the "Interactivity" and the "Real-time Response Speed". Therefore, the system successfully provides comprehensive, thorough, and correct learning content and information. Most students find the system's interface intuitive and easy to use. Therefore, our OpenCHM demonstrates strong performance in its core informational functions but enhance the system's technical Interactivity and responsiveness.

Fig. 15 presents the quantitative results. Our system approach significantly outperforms the traditional method for Searching by Herb Name, Conditional Filtering and Chinese Herbal Medicine Q&A tasks. Both the absorption score and the study's time for our system group is significantly than that of the traditional group. All tasks received scores below 80 by the traditional method while these tasks can get scores above 80 by our system approach. In particular, student in the first group spend about 120 seconds reading textbooks for questions related to Chinese Herbal Medicine Q&A. In contrast, students in the second group can only take about 90 seconds by our OpenCHM system for Chinese Herbal Medicine Q&A. This suggests that our system learning method effectively increases students' immersion and effectiveness in the educational process.

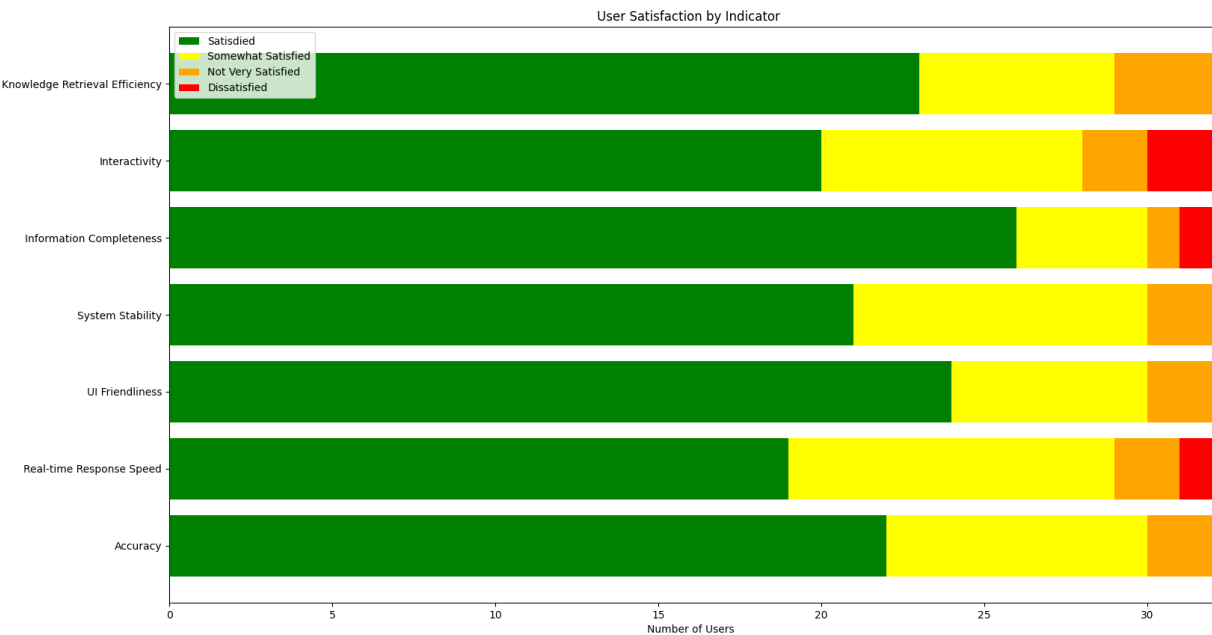


Figure 14: User Satisfaction with our OpenCHM learning experience. There are 4 indicators to evaluate the satisfaction levels such as Satisfied, Somewhat Satisfied, Not Very Satisfied and Dissatisfied.

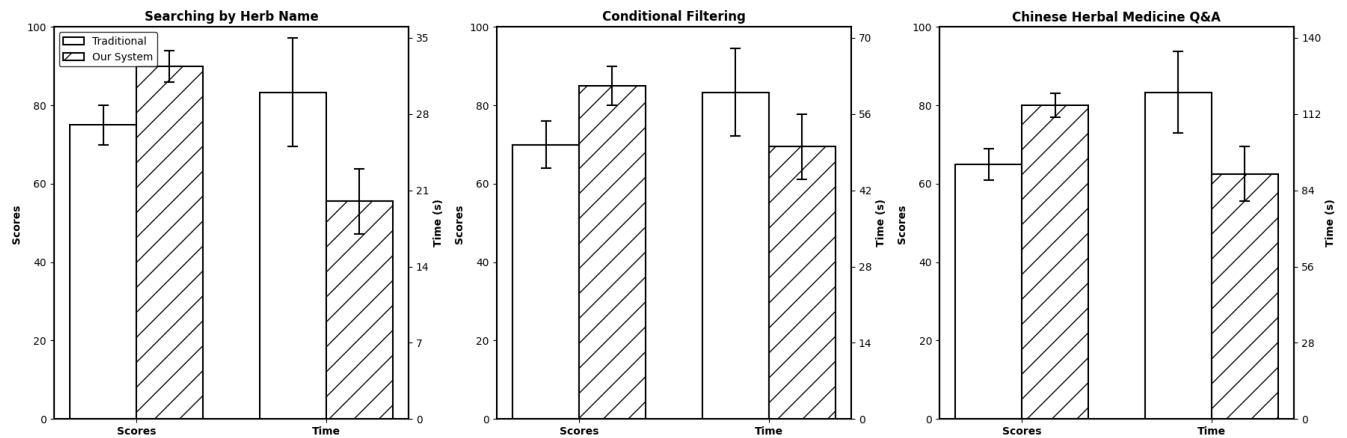


Figure 15: Scores of each Q&A task. The first group with a traditional learning method (Traditional) vs the second group with the OpenCHM learning method (Our System).

Our learning approach makes them more deeply interested and engaged.

5 Conclusion

We present OpenCHM, an open-source MKG-based education system for CHM. Our MKG integrates texts and images, which can give students a comprehensive perception of CHM. OpenCHM makes

a difference in several key applications such as knowledge visualization, multimodal knowledge retrieval and a Q&A platform. A variety of educational examples are provided for students to study concepts of CHM and promote them to explore further information. In conclusion, OpenCHM enables to achieve alignment between teaching and learning when providing smart tools to advance CHM education. In the future, our work will focus on expanding the scale

of the knowledge graph of CHM, such as more various entity and relationships between entities. Additionally, we aim to improve performance of OpenCHM through integrating more multimodal data and supporting a wider range of learning objectives. Finally, OpenCHM is combined with healthcare, which gives users tailored advices based on their health information.

Acknowledgments

This work is supported by the Shaanxi Province Key Research and Development Program (Grant Number S2024-YF-ZDXM-CYSL-0106).

References

- Xiang Chen, Ningyu Zhang, Lei Li, Shumin Deng, Chuanqi Tan, Changliang Xu, Fei Huang, Luo Si, and Huajun Chen. 2022. Hybrid transformer with multi-level fusion for multimodal knowledge graph completion. In *Proceedings of the 45th international ACM SIGIR conference on research and development in information retrieval*. 904–915.
- Zhuo Chen, Yichi Zhang, Yin Fang, Yuxia Geng, Lingbing Guo, Jiaoyan Chen, Xiaozhe Liu, Jeff Z Pan, Ningyu Zhang, Huajun Chen, et al. 2025. Knowledge Graphs for Multi-modal Learning: Survey and Perspective. *Information Fusion* (2025), 103124.
- Christine Dewi, Rasoul Amirzadeh, Dhananjay Thiruvady, and Nayyar Zaidi. 2025. Knowledge-Guided Object Detection via Bayesian Networks and Knowledge Graphs (KGBNCNet). *Expert Systems with Applications* (2025), 129385.
- Sebastián Ferrada, Benjamin Bustos, and Aidan Hogan. 2017. IMGpedia: a linked dataset with content-based analysis of Wikimedia images. In *International Semantic Web Conference*. Springer, 84–93.
- Jinglin He, Yunqi Guo, Lai Kwan Lam, Waikei Leung, Lixing He, Yuanan Jiang, Chi Chiu Wang, Guoliang Xing, and Hongkai Chen. 2025. Opentcm: a graphrag-empowered llm-based system for traditional chinese medicine knowledge retrieval and diagnosis. *arXiv preprint arXiv:2504.20118* (2025).
- Xiao Ke, Baitao Chen, Xiong Yang, Yuhang Cai, Hao Liu, and Wenzhong Guo. 2025. Cross-modal independent matching network for image-text retrieval. *Pattern Recognition* 159 (2025), 111096.
- Weisheng Li, Hao Zhong, Junming Zhou, Chao Chang, Ronghua Lin, and Yong Tang. 2024. An attention mechanism and residual network based knowledge graph-enhanced recommender system. *Knowledge-Based Systems* 299 (2024), 112042.
- Ye Liu, Hui Li, Alberto Garcia-Duran, Mathias Niepert, Daniel Onoro-Rubio, and David S Rosenblum. 2019. MMKG: multi-modal knowledge graphs. In *European Semantic Web Conference*. Springer, 459–474.
- Junjie Long, Jinghao Niu, Jiaxi Liu, Jie Li, Wensheng Zhang, et al. 2024. RTGN: Robust Traditional Chinese Medicine Graph Networks for Patient Similarity Learning. *IEEE Journal of Biomedical and Health Informatics* (2024).
- Xiang Shen, Chongqing Chen, Dezhi Han, Yangshuyi Xu, Xiuying Wang, and Huiyu Zhou. 2025. A Triple-Branch Hybrid Dynamic-Static Alignment Strategy for Vision-Language Tasks. *Neural Networks* (2025), 107871.
- Jingjing Wang, Haoran Xie, Siyu Zhang, S Joe Qin, Xiaohui Tao, Fu Lee Wang, and Xiaoliang Xu. 2025. Multimodal fusion framework based on knowledge graph for personalized recommendation. *Expert Systems with Applications* 268 (2025), 126308.
- Xin Wang, Benyuan Meng, Hong Chen, Yuan Meng, Ke Lv, and Wenwu Zhu. 2023. TIVA-KG: A multimodal knowledge graph with text, image, video and audio. In *Proceedings of the 31st ACM international conference on multimedia*. 2391–2399.
- Yang Wu, Feilong Zhang, Kuo Yang, Shuangfang Fang, Dechao Bu, Hui Li, Liang Sun, Hairuo Hu, Kuo Gao, Wei Wang, et al. 2019. SymMap: an integrative database of traditional Chinese medicine enhanced by symptom mapping. *Nucleic acids research* 47, D1 (2019), D1110–D1117.
- Bo Zhang, Ruifang Li, Kedong Yin, Shuo Hua, Shiyu Li, Mengwan Jiang, Haoping An, and Peng Li. 2025. Construction and Application of Traditional Chinese Medicine Knowledge Graph Based on Large Language Model. *Interdisciplinary Sciences: Computational Life Sciences* (2025), 1–13.